

Amadou MAMANE
Senior AI Engineer | GenAI & ML Systems (GCP)
CentraleSupélec Alumni

AI/ML Engineer with 10 years of experience shipping ML and Generative AI systems into production across banking, retail, and telecommunications environments. I bridge research and production, owning the full journey from architecture decisions to scalable, observable systems on cloud-native and hybrid enterprise environments. Comfortable operating in both technical and business contexts, from early scoping to long-term operations.

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MAIN ACHIEVEMENTS

- **Productionized a multi-agent GenAI platform on GCP** (LangGraph + Vertex AI) for a major German retailer with 950 stores, serving around 1,000 daily users.
- **Deployed real-time fraud detection model** on a payment network processing roughly 1.5 billion transactions per day.
- **Built and deployed churn-scoring models across multiple countries**, scoring tens of millions of banking customers end-to-end on GCP.
- **Google Cloud certified** — Professional ML Engineer, Professional Cloud Architect, Generative AI Leader.

SKILLS

- **GenAI & LLMs** RAG, multi-agent systems, prompt engineering, evaluation, hallucination mitigation • Models: Gemini, GPT, Llama, Claude, Gemma, DeepSeek • Frameworks: LangChain, LangGraph, LangSmith, Google ADK • Vector databases: AlloyDB, Qdrant, Pinecone
- **ML / DL** Forecasting, clustering, recommendation, NLP, computer vision, graph analytics, time series • TensorFlow, PyTorch, Scikit-learn, Spark ML, Statsmodels
- **Data & Cloud** GCP (Vertex AI, BigQuery, Firestore, Dataflow, AlloyDB, Dataproc, Cloud Run, Pub/Sub, Composer) • AWS, OpenStack • Hadoop, Spark, Flink, Kafka, Hive, Pig, NiFi, distributed pipelines, Redis, Cassandra, MySQL
- **MLOps & LLMOps** CI/CD, model monitoring, Observability • Git, Terraform, Airflow, Kubeflow, Docker, Kubernetes • OpenTelemetry, Cloud Trace/Logging/Monitoring, Grafana
- **Programming** Python, SQL, Scala, Java, Shell

CERTIFICATIONS & LANGUAGES

- Google Cloud Certified - Generative AI Leader (September 2025)
- Google Cloud Certified - Professional Machine Learning Engineer (October 2025)
- Google Cloud Certified - Professional Cloud Architect (April 2026)
- MIT - Applied Data Science Program (July 2024)
- French - Fluent • English - Professional working proficiency

EDUCATION

MIT — Applied Data Science Program: Leveraging AI for Effective Decision Making, 2024
École Centrale Paris (CentraleSupélec) — M.S. in Computer Science, 2015
École des Mines de Saint-Étienne — M.S.(Research) in Industrial Engineering & Computer Science, 2012

EXPERIENCE

June 2025 – Present

Thalia (via Devoteam GCloud), Germany

Senior AI Engineer

Designing and shipping a **production multi-agent GenAI platform** for an omnichannel book retailer (online + 950 stores), serving around **1,000 daily users and tens of thousands of daily requests**.

- Built a multi-agent orchestration system on **LangGraph and Vertex AI**, enabling complex customer-support and product-discovery flows in a single conversational interface.
- Owned the end-to-end data/AI architecture on GCP, including the trade-off analysis between **AlloyDB vector search and BigQuery ML** for multimodal (text + image) retrieval — selected based on latency, cost, and operational fit.
- Industrialized the platform with **GitLab CI/CD and Terraform**, and instrumented it end-to-end with **OpenTelemetry, Cloud Trace, Cloud Logging, and Cloud Monitoring** — giving the team full traceability across agent calls, retrieval steps, and LLM invocations.
- Defined reusable AI components and engineering standards adopted across the team to ensure maintainability and production-readiness.

July 2024 – May 2025

Applied Research Project (MIT), Remote

AI Engineer

Built an end-to-end **Retrieval-Augmented Generation** system to query heterogeneous enterprise sources (PDF, CSV, HTML, SQL) in natural language.

- Designed a multi-format ingestion pipeline with robust chunking strategies (token-size fallback) and metadata enrichment to power hybrid search (vector + lexical) on **Qdrant**.
- Implemented contextual reranking and a confidence-scoring layer on top of open-source LLMs (**LLaMA, Gemma**) to mitigate hallucinations and enforce source citation in answers.
- Improved answer precision by **~30% over a vanilla dense-retrieval baseline** in qualitative evaluation, with structured, source-attributed outputs.

September 2023 – November 2024

Cartes Bancaires, Paris

Senior Data scientist

Member of the anti-fraud team protecting the French interbank payment network — around **1.5B transactions per day**.

- Co-developed a **Gradient Boosting model (Spark ML)** to proactively detect attacks targeting e-merchants, optimized for extremely rare-signal detection on massive transactional data, deployed in production on around 1.5B transactions/day.
- Industrialized the full pipeline with **PySpark and Oozie**, and authored Spark best-practice guidelines adopted across the team.

April 2019 – August 2023

RCI Bank & Services (Renault Group), Paris

Senior Data scientist

Core contributor to the group's **customer data platform** and strategic ML **applications**, from on-prem Hadoop through the full migration to **GCP**.

- Built a **graph-based customer deduplication engine** combining vector similarity and connected-components analysis on tens of millions of cross-referenced records, resolving entity clusters of up to 1,000+ records for corporate clients and materially improving downstream data quality.
- Designed and shipped a **churn prediction score** running across the group's customer base in multiple countries — including nearly 18M customers in France, 15M in Germany, 11M in the UK, among others — feeding CRM and retention campaigns.
- Industrialized ML workflows with **GitLab CI/CD, Airflow, and Kubeflow** on GCP (Vertex AI,

BigQuery, Dataproc, Pub/Sub, Cloud Run, Composer).

- Led several POCs to evaluate and integrate emerging data/ML technologies into the platform roadmap.

May 2017 – March 2019

Orange (via Altran), Arcueil

Senior ML engineer

Real-time processing of mobile network data (2G/3G/4G) feeding business applications across the group.

- Built streaming applications on **Kafka, Flink, and Spark** to normalize and broadcast network events to downstream business teams.
- End-to-end ownership— prototyping, testing, component selection, production deployment, monitoring (**InfluxDB, Grafana**), and maintenance of business-critical services.

EARLIER ROLES

Data Scientist — SNCF (via Altran), 2016–2017: Yield-management decision-support tools — train-occupancy and price-sensitivity models in PySpark.

Data Scientist — Orange (via Altran), 2015–2016: Implementation of Nokia's CEMOD platform for customer-experience and network-quality scoring.

Data Scientist Intern — Société Générale, 2015: Innovative DDoS detection from web access logs (vs. traditional network-volume approaches), using clustering + supervised classification.